

Sampling Distributions and the Central Limit Theorem

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Population, sample, parameter and statistic ([Note 220](#)); Normal distribution ([Note 240](#)).
- **Leads to:** Confidence intervals ([Note 280](#)); Hypothesis testing: null and alternative ([Note 290](#)); Z-test and rejection regions ([Note 291](#)).

1. Overview

Key point: Take many samples of the same size, compute each sample's mean, and the means form a normal distribution centred on the population mean, whatever the shape of the population.

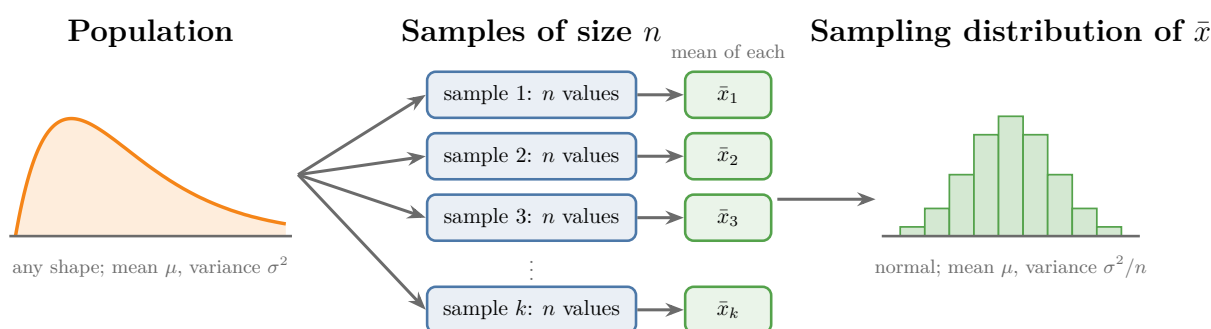


Figure 1: From a population of any shape, through many samples, to the sampling distribution of the mean

Figure 1 shows the process this Note is built on. We draw many samples of size n from a population, compute one number per sample (here the mean), and look at the distribution of those numbers. That distribution is a **sampling distribution** (G-1736).

The **central limit theorem (CLT)** (G-364) says what this distribution looks like for the mean: a normal curve with the population's mean and a spread that shrinks as n grows. The CLT holds for skewed, flat and even discrete populations, and it is the base of the next topics, confidence intervals and hypothesis testing.

This Note covers:

- the sampling distribution of a statistic;
- the statement of the CLT and its conditions;
- the mean μ and variance σ^2/n of the sample means, and the standard error;
- simulations from five non-normal populations;
- why the CLT matters.

2. Population and sample, in brief

Key point: We want a number about the whole population, but we can only measure samples.

Recap (see the [what is statistics Note](#)): the **population** (G-1525) is the entire group we want to study; a **sample** (G-1731) is the part we actually measure. A number computed from the population, such as μ , is a **parameter** (G-1448); the same number computed from a sample, such as $\bar{x}$, is a **statistic** (G-1880).

Our running example is the average monthly salary in India. The population is the salary of all 140 crore people. Asking all of them is impossible, so we ask, say, 50,000 people chosen at random from every state and district, and infer the national average from them.

3. Sampling distributions

Key point: A sampling distribution is the distribution of a statistic (a mean, a variance, a proportion) computed from many independent samples of the same size from one population.

The salaries of the whole population are a set of numbers, so they follow some distribution. Most likely a skewed one: many people earn little and a few earn a lot, like the orange curve of Figure 1. We do not know its exact shape.

Now we build a sampling distribution in four steps:

1. Choose a **sample size** (G-1727), say $n = 50$ people.
2. Draw 50 people at random and record their salaries: sample 1.
3. Repeat, drawing a new random sample of 50 each time, until we have, say, 100 samples: 100 sets of 50 numbers.
4. Compute the mean of every sample: $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_{100}$.

These 100 sample means form the **sampling distribution of the sample mean**. Two sizes must not be confused: the sample size is $n = 50$ (people per sample); the number of samples is 100.

The statistic need not be the mean. If we compute the variance of each of the 100 samples instead, the 100 sample variances form the **sampling distribution of the sample variance**. The same works for the standard deviation, the coefficient of variation or a proportion.

1. **In words:** a sampling distribution is the distribution of a statistic computed from many independent samples of the same size from one population.
2. **Formula:** for the mean, from k samples of size n ,

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}, \quad j = 1, \dots, k$$

and the sampling distribution is the distribution of $\bar{x}_1, \dots, \bar{x}_k$.

3. **Example:** three samples of size $n = 3$ give means

$$\bar{x}_1 = \frac{2 + 4 + 9}{3} = 5, \quad \bar{x}_2 = \frac{1 + 3 + 5}{3} = 3, \quad \bar{x}_3 = \frac{6 + 7 + 8}{3} = 7$$

The numbers 5, 3 and 7 are three points of the sampling distribution of the mean.

Figure 2 draws this example. Watch each orange mark: it sits at the balance point of its sample's three blue values, and only the marks drop to the bottom line, where the sampling distribution is built.

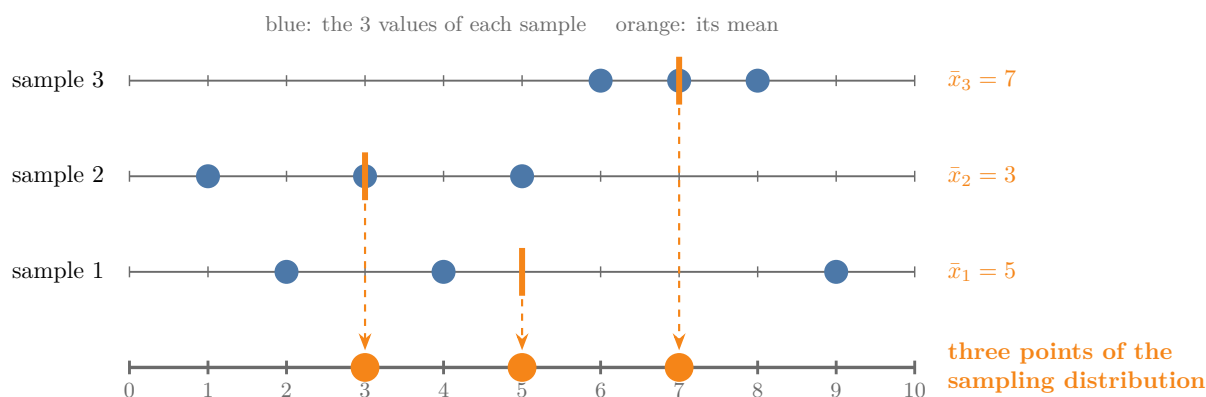


Figure 2: Three samples of size 3 (blue values) and their means 5, 3 and 7 (orange), which become three points of the sampling distribution

3.1 Why sampling distributions matter

Key point: A sampling distribution shows how much a statistic varies from sample to sample, and that variability is what inference needs.

Every sample of 50 people gives a slightly different mean. The sampling distribution measures exactly how different: it shows the **variability of a sample statistic**. Knowing that variability lets us:

- compute confidence intervals for a population parameter;
- perform hypothesis tests;
- predict a population parameter, such as the mean, from sample data.

The first two are the topics of later Notes. The third is the subject of this Note and the next one.

4. The central limit theorem

Key point: The sample means of large independent samples follow a normal distribution, regardless of the distribution of the population.

1. **In words:** the distribution of the sample means of many independent and identically distributed values approaches a normal distribution, regardless of the distribution of the values themselves.
2. **Formula:** for samples of size n from a population with mean μ and variance σ^2 ,

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{for large } n$$

3. **Example:** an exponential population has $\mu = 1$ and $\sigma^2 = 1$. For samples of $n = 30$, the CLT predicts that the sample means follow

$$N\left(1, \frac{1}{30}\right) = N(1, 0.0333), \quad \text{standard deviation } \sqrt{0.0333} = 0.183$$

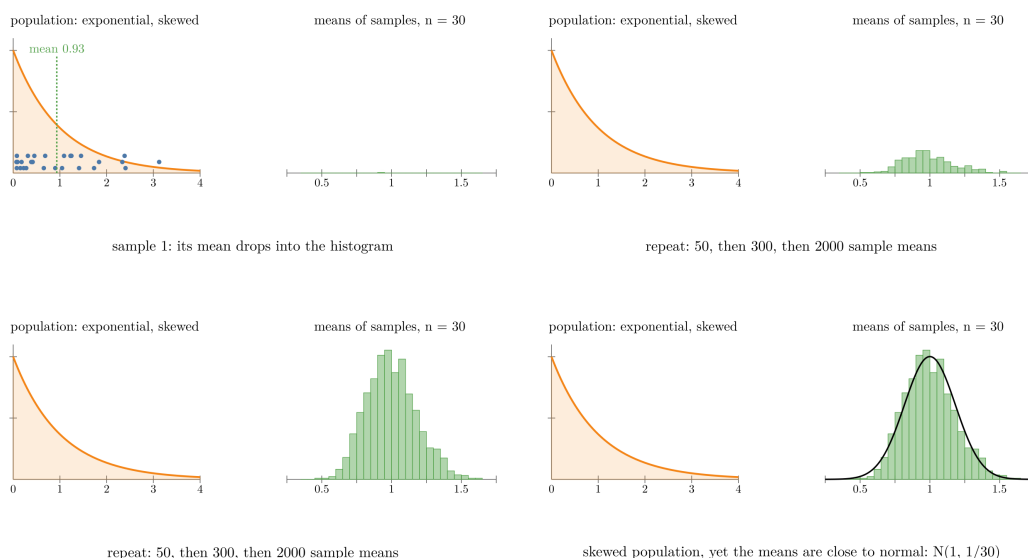


Figure 3: Samples of 30 from a skewed exponential population: each sample's mean drops into the histogram, which grows into the normal curve $N(1, 1/30)$

Figure 3 runs this experiment. The population on the left is strongly skewed: most values are near 0 and a few are large. Each sample of 30 values gives one mean, which drops into the histogram on the right. After 2000 samples, the histogram is close to the bell of $N(1, 1/30)$.

In the salary example: draw 100 people, record the mean salary, repeat 1000 times, and plot the 1000 means. The CLT says the plot is a normal curve. The shape of the salaries themselves does not matter. They may be:

- log-normal;
- uniform (every salary in a range equally likely);
- Pareto;
- binomial;

- or no named distribution at all.

An everyday picture: one die is flat, every face from 1 to 6 equally likely. The average of 30 dice, though, almost never lands near 1 or 6, because that needs nearly every die to be low, or nearly every die to be high. Most averages land near 3.5, and the further from 3.5, the rarer they get: a bell.

Figure 4 tests this with four very different dice: a fair one, one that mostly rolls low, a U-shaped one that mostly rolls 1 or 6, and one that mostly rolls 6. For each die we compute the exact distribution of the sum of n rolls. The sums drift to the right and spread out as n grows, so to compare their shapes we re-centre and re-scale each one: subtract its mean $n\mu$ and divide by its standard deviation $\sqrt{n}\sigma$ (section 4.2). Watch the bottom row: at $n = 1$ the four shapes are as different as the dice; by $n = 10$ they are close to one bell; at $n = 50$ all four sit on the same curve, the **standard normal distribution** $N(0, 1)$ (G-1873). Re-scaling the mean instead of the sum gives exactly the same picture, since the mean is the sum divided by n .

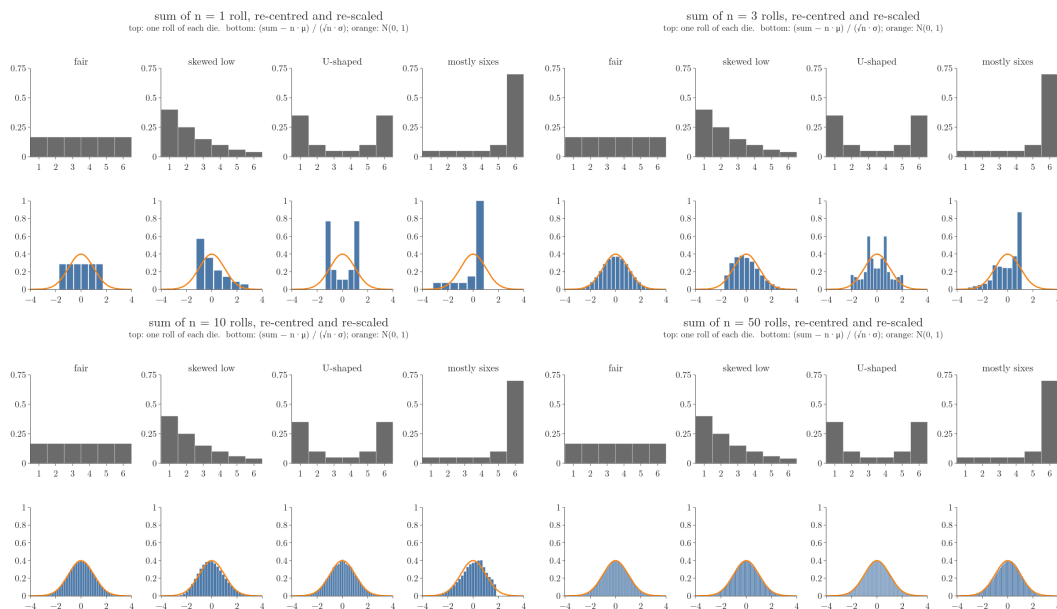


Figure 4: Four different dice (top) and the re-centred, re-scaled sum of n rolls of each (bottom) for $n = 1$ to 50: all four approach the same normal curve (idea after 3Blue1Brown, “But what is the Central Limit Theorem?”)

The CLT makes no claim about the shape of the population or of a single sample. A sample of 100 salaries still looks skewed; only the **means** of many samples form a bell.

Extra: The CLT is a proved theorem, not just an observed fact of nature: Laplace proved a first version in 1810 and Lyapunov a general one in 1901 (Fischer 2011). The intuition: a sample mean adds up many independent pieces, so unusually large and unusually small values tend to cancel, and the sum forgets the shape of each piece. Among shapes with a finite variance, the normal curve is the only one that stays the same when we add copies

and rescale (Feller 1971, §VI.1).

4.1 Conditions

Key point: The CLT needs a large enough sample (usually $n \geq 30$), a population with finite variance, and independent, identically distributed values.

1. **Large enough sample size.** The usual rule is $n \geq 30$, a rule of thumb rather than a sharp limit. How many values are enough depends on the population's shape: a symmetric population gives bell-shaped means already at small n , while a very skewed one needs more (see the Extra box on how fast the means become normal, and section 5.1).
2. **Finite variance.** The population must have a finite variance. Every finite population has one; a theoretical infinite population, such as a Pareto distribution with a small α , may not (see the Extra box on infinite variance below).
3. **Independent and identically distributed (i.i.d.) values.** **Independent:** one value does not affect another. **Identically distributed:** every value comes from the same population, so each has the same distribution. Drawing at random from one population gives both.

Extra: How fast the means become normal. For independent values the cumulants of a sum add up, just like the variance (the second cumulant). The r -th cumulant of $\bar{X}$ is therefore $n\kappa_r/n^r$, and dividing the third by $\sigma_{\bar{x}}^3 = \sigma^3/n^{3/2}$ and the fourth by $\sigma_{\bar{x}}^4 = \sigma^4/n^2$ gives

$$\text{skewness of } \bar{X} = \frac{\gamma_1}{\sqrt{n}}, \quad \text{excess kurtosis of } \bar{X} = \frac{\gamma_2}{n}$$

where γ_1 and γ_2 are the population's skewness and excess kurtosis. A symmetric population ($\gamma_1 = 0$) gives symmetric means at every n : for the uniform population ($\gamma_2 = -1.2$), $n = 5$ already gives skewness 0.02 and excess kurtosis -0.24 in the Notebook, matching $-1.2/5$. A very skewed population needs more: the exponential ($\gamma_1 = 2$) still has skewness $2/\sqrt{30} = 0.37$ at $n = 30$ (section 5.1).

Extra: When the variance is infinite, the CLT fails. A Pareto distribution with $\alpha = 1.5$ has mean 3 but infinite variance (see the [Pareto Note](#)). In 5000 simulated samples, the skewness of the sample means is 17.8 for $n = 30$ and still 19.5 for $n = 300$: no bell appears. For such tails the generalised CLT says the scaled sums approach a skewed “stable” distribution, not a normal one (Gnedenko and Kolmogorov 1954). The Notebook also shows how one value can dominate: in samples of 300, the largest single value makes up at least 55% of the sample's total in 1% of the Pareto samples, against 3.4% at the same point for the exponential population.

4.2 Mean and variance of the sample means

Key point: The sample means are centred on the population mean μ , and their variance is σ^2/n ; their standard deviation, $\sigma/\sqrt{n}$, is the standard error.

The CLT gives more than the shape. If the population has mean μ and variance σ^2 :

- the mean of the sampling distribution is exactly μ ;
- its variance is σ^2/n , where n is the **sample size** (not the number of samples).

The standard deviation of the sampling distribution has its own name, the **standard error** (G-1872) of the mean.

1. **In words:** the standard error is the population standard deviation divided by the square root of the sample size. Because the variance is divided by n , the standard deviation is divided by $\sqrt{n}$.

2. **Formula:**

$$SE = \sigma_{\bar{x}} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$$

3. **Example:** a gamma population with mean 2 and variance 2 (so $\sigma = 1.414$), samples of $n = 50$:

$$\sigma_{\bar{x}}^2 = \frac{2}{50} = 0.04, \quad SE = \frac{1.414}{\sqrt{50}} = \frac{1.414}{7.071} = 0.2$$

The sample means vary around 2 with a standard deviation of only 0.2, while single values vary with a standard deviation of 1.414.

Why the square root. When we add n independent values, the **variances add**, not the standard deviations. So the sum has variance $n\sigma^2$ and standard deviation $\sqrt{n}\sigma$: the spread of a sum grows, but only as $\sqrt{n}$. Dividing the sum by n to get the mean divides that spread by n , which leaves $\sqrt{n}\sigma/n = \sigma/\sqrt{n}$ (the Extra below writes it out).

A second example: 100 dice. One fair die has $\mu = 3.5$ and $\sigma = 1.71$.

1. **The sum of 100 dice** has mean $100 \times 3.5 = 350$ and standard deviation $\sqrt{100} \times 1.71 = 17.1$.
2. **By the CLT the sum is about normal**, so 95% of sums land within 2 standard deviations: $350 \pm 2 \times 17.1$, from 316 to 384.
3. **Dividing the sum by 100 gives the average of the 100 dice:** mean 3.5, standard error $1.71/\sqrt{100} = 0.171$, and 95% of averages land between 3.16 and 3.84.

A larger sample gives a smaller standard error: four times the sample size halves it. So sample means from big samples cluster tightly around μ .

Extra: Why the variance is σ^2/n . For independent values, the variance of a sum is the sum of the variances, so $x_1 + \dots + x_n$ has variance $n\sigma^2$. Dividing by n to get the mean divides the variance by n^2 :

$$\text{Var}(\bar{X}) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

This completes the argument of the [measures of dispersion Note](#) for Bessel's correction. There, measuring distances from $\bar{x}$ instead of μ lost the amount $n(\bar{x} - \mu)^2$. Its average is $n \cdot E[(\bar{x} - \mu)^2] = n \cdot \sigma^2/n = \sigma^2$: exactly one σ^2 . So the sum of squared distances from $\bar{x}$ averages $(n - 1)\sigma^2$, and dividing by $n - 1$ is right on average.

5. Simulating the CLT

Key point: Simulations from uniform, exponential, Poisson, gamma and binomial populations all give bell-shaped sample means with the mean and spread the CLT predicts.

We test the theorem by simulation. We pretend a known distribution is the population, draw many samples from it, and plot the sample means.

Python: A sampling distribution in three lines.

```
import numpy as np

rng = np.random.default_rng(42)      # fixed seed
samples = rng.uniform(0, 1, size=(1000, 30))
sample_means = samples.mean(axis=1)
samples has shape (1000, 30): 1000 samples (rows) of size 30 (columns).
.mean(axis=1) takes the mean of each row, giving 1000 sample means.
```

For a **uniform** population on $[0, 1]$ (see the [uniform and log-normal Note](#)), every value is equally likely: the population is flat, not bell-shaped. Yet the 1000 sample means of size 30 form a bell centred at 0.50, with standard deviation 0.052, matching the CLT's $\sqrt{(1/12)/30} = 0.053$. With $n = 300$ the bell is narrower: 0.0166, against the predicted 0.0167.

The other populations behave the same way (Figure 5):

- **Exponential:** waiting times between random events; strongly right-skewed, $\mu = \sigma = 1$.
- **Poisson:** counts of events in a fixed interval (see the [PDF Note](#)); here with mean 3.
- **Gamma:** a right-skewed continuous distribution; here with mean 2 and variance 2.
- **Binomial:** successes in 10 trials with $p = 0.2$ (see the [Bernoulli and binomial Note](#)).

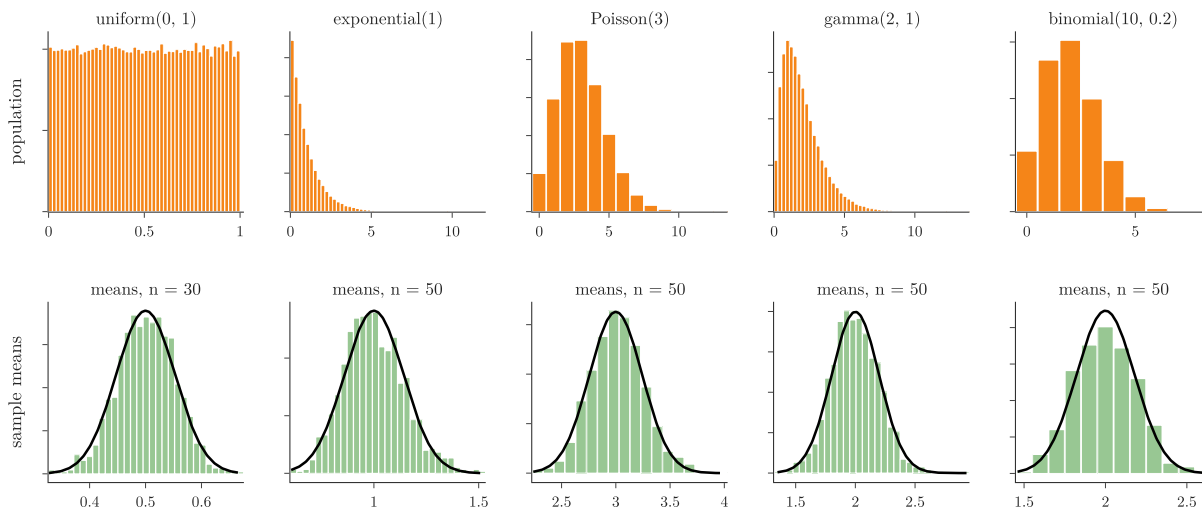


Figure 5: Top: five non-normal populations. Bottom: the means of 1000 samples from each, with the normal curve the CLT predicts

Every population in the top row has a different shape: flat, skewed, discrete. Every histogram in the bottom row is a bell that follows the black normal curve $N(\mu, \sigma^2/n)$.

Population	Skewness of population	Mean of sample means	Std of sample means	CLT: $\sigma/\sqrt{50}$	Skewness of means
exponential(1)	2.00	0.998	0.143	0.141	0.26
Poisson(3)	0.58	2.993	0.249	0.245	0.06
gamma(2, 1)	1.41	2.002	0.195	0.200	0.13
binomial(10, 0.2)	0.47	1.996	0.183	0.179	0.05

The skewness drops from up to 2.00 in the populations to at most 0.26 in the sample means (see the [skewness Note](#)).

5.1 The effect of the sample size

Key point: As n grows, the sampling distribution becomes more normal and narrower; its standard deviation follows $\sigma/\sqrt{n}$.

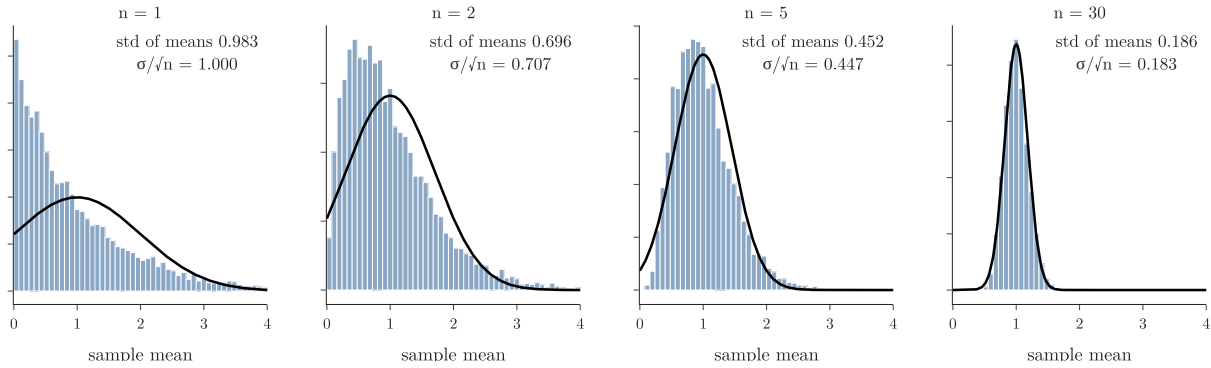


Figure 6: Sample means of an exponential population for $n = 1, 2, 5$ and 30 (10,000 samples each), with the curve $N(1, 1/n)$

Figure 6 uses the exponential population, the most skewed one:

n	Std of sample means	$\sigma/\sqrt{n}$	Skewness of means
1	0.983	1.000	1.93
2	0.696	0.707	1.34
5	0.452	0.447	0.90
30	0.186	0.183	0.38

With $n = 1$, each “mean” is a single value, so the histogram is the population itself. As n grows, the histogram narrows as $\sigma/\sqrt{n}$ predicts, and the skewness falls as $2/\sqrt{n}$ predicts (section 4.1): 2.00, 1.41, 0.89 and 0.37. At $n = 30$ a small skew of 0.38 remains: for very skewed populations, the rule $n \geq 30$ gives a shape that is close to normal, not perfectly normal.

6. Checking the mean and variance

Key point: For a gamma population with $\mu = 2$ and $\sigma^2 = 2$, 10,000 sample means of size 50 have mean 1.997 and variance 0.0405, against the predicted 2 and 0.04.

To check the second part of the CLT, we use a population whose mean and variance we know exactly: gamma with shape 2 and scale 1, so $\mu = 2$ and $\sigma^2 = 2$. We draw 10,000 samples of size 50.

Quantity	CLT prediction	Simulation
Mean of the sample means	$\mu = 2$	1.9974
Variance of the sample means	$\sigma^2/n = 2/50 = 0.04$	0.0405
Standard error	$\sigma/\sqrt{n} = 0.2$	0.2012

Both properties hold: the sample means are centred on the population mean, and their variance is the population variance divided by n .

Figure 7 replays the check as the samples come in. Watch both curves: after a few samples they swing widely, and after a few hundred they settle on the dashed lines the CLT predicts, 2 and 0.04.

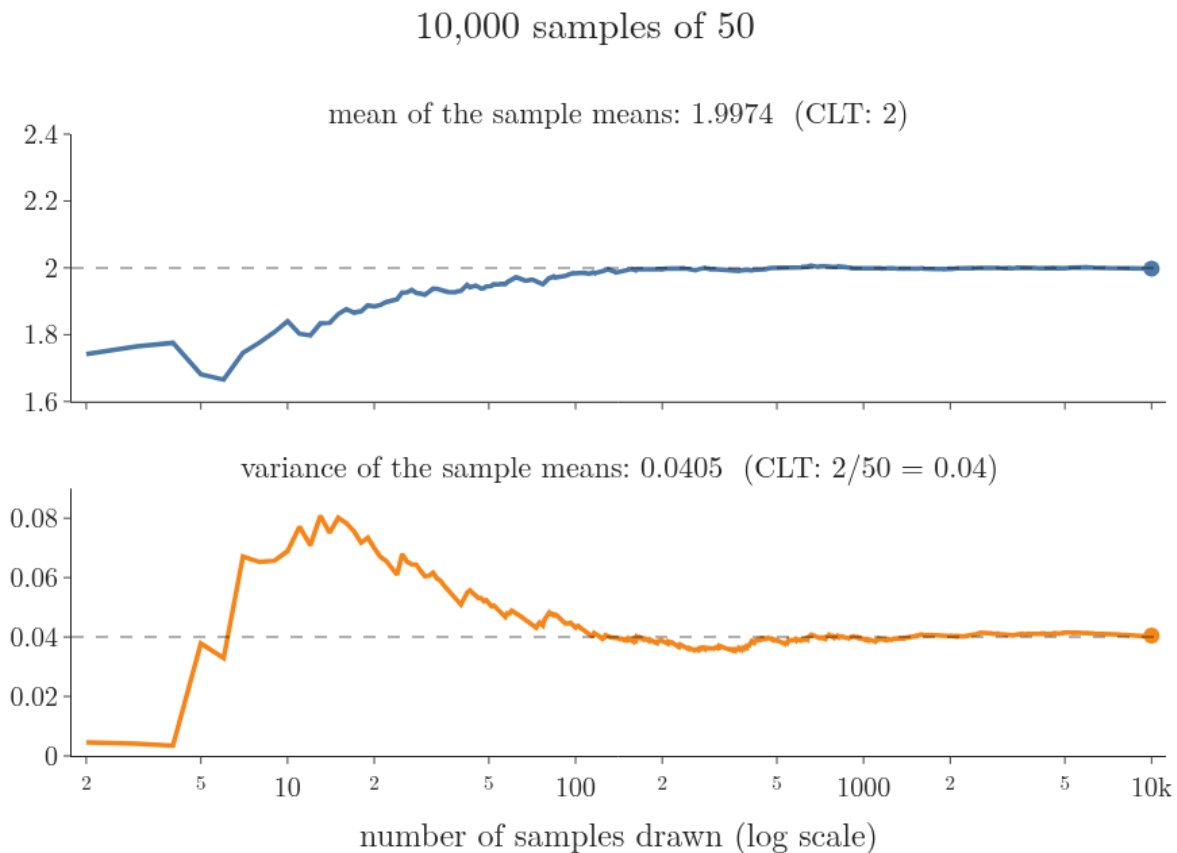


Figure 7: The mean (top) and variance (bottom) of the sample means as 1 to 10,000 samples of 50 accumulate from $\text{gamma}(2, 1)$; dashed: the CLT values 2 and 0.04

Python: The check in code.

```
from scipy import stats
```

```
gamma = stats.gamma(2)           # shape 2, scale 1
gamma.mean(), gamma.var()        # 2.0, 2.0
rng = np.random.default_rng(7)
means = rng.gamma(2, 1, size=(10_000, 50)).mean(axis=1)
means.mean(), means.var()        # 1.9974, 0.0405
```

7. Why the CLT matters

Key point: The CLT lets us infer a population mean from samples without knowing the population's distribution, and it justifies confidence intervals, hypothesis tests, t-tests, ANOVA and linear regression.

The two parts of the theorem turn into a method. We do not know the salary distribution of 140 crore people, but we know that sample means are normal around the true mean μ , with standard error $\sigma/\sqrt{n}$. So a normal curve built from sample data tells us where μ must lie (see the [standard normal Note](#)). In principle, the variance of the sample means also reveals the population variance: $\sigma^2 = n \times \text{Var}(\bar{X})$.

The same works anywhere. A t-shirt company that does not know the heights of all its customers can sample a few thousand of them and estimate the average height of all of them. The [next Note](#) does this on real data.

The CLT also justifies many standard techniques:

- **confidence intervals:** a range for a population parameter;
- **hypothesis tests**, such as the t-test and ANOVA;
- **linear regression:** its tests on the coefficients rely on normality, which the CLT provides for large samples (Wooldridge, ch. 5).

The CLT has one big condition hidden in the word “random”: the samples must be random and representative (see sampling bias in the [challenges in ML Note](#)). Salaries collected only in cities give a sampling distribution centred on the average city salary, not on India's average. The CLT cannot fix a biased sample.

8. Summary

Idea	Formula	Example
Sampling distribution of the mean	distribution of $\bar{x}_1, \dots, \bar{x}_k$	100 means of samples of 50 salaries
CLT, shape	$\bar{X} \approx N(\mu, \sigma^2/n)$	exponential, $n = 30$: $N(1, 1/30)$
Mean of the sample means	μ	gamma: 1.9974 against 2
Variance of the sample means	σ^2/n	$2/50 = 0.04$; simulated 0.0405
Standard error	$\sigma/\sqrt{n}$	$1.414/\sqrt{50} = 0.2$

- A sampling distribution comes from many samples of the **same size**; do not confuse the sample size n with the number of samples.
- The CLT holds for any population with finite variance, given i.i.d. values and a large enough n (usually $n \geq 30$).
- Larger samples give narrower, more normal sampling distributions.
- The CLT does not fix biased samples: they must be random and representative.

9. Sources

Built from

- CampusX, “Session 43 - Central Limit Theorem | DSMP 2023”, YouTube, <https://www.youtube.com/watch?v=-WmJDYBor7c>
- StatQuest with Josh Starmer, “The Central Limit Theorem, Clearly Explained!!!”, YouTube, <https://www.youtube.com/watch?v=YAlJCEDH2uY>
- Khan Academy, “Sampling distribution of the sample mean”, YouTube, <https://www.youtube.com/watch?v=KE>
- 3Blue1Brown, “But what is the Central Limit Theorem?”, YouTube, <https://www.youtube.com/watch?v=KE>

Other references

- Feller, W. (1971). *An Introduction to Probability Theory and Its Applications*, Vol. II, 2nd ed. Wiley. §VI.1.
- Fischer, H. (2011). *A History of the Central Limit Theorem: From Classical to Modern Probability Theory*. Springer.
- Gnedenko, B. V. and Kolmogorov, A. N. (1954). *Limit Distributions for Sums of Independent Random Variables*. Addison-Wesley.
- Wooldridge, J. M. *Introductory Econometrics: A Modern Approach*. Cengage. Chapter 5, “Multiple Regression Analysis: OLS Asymptotics”.

10. Key terms

Term	Meaning
Sampling distribution	The distribution of a statistic computed from many independent samples of the same size from one population
Sampling distribution of the sample mean	The distribution of the means of many samples of size n
Sample size (n)	The number of values in one sample
Central limit theorem	Averages of samples follow a normal distribution, whatever the distribution of the data; their mean is μ and their variance σ^2/n
Standard error	The standard deviation of a sampling distribution; for the mean, $\sigma/\sqrt{n}$
Independent and identically distributed (i.i.d.)	Values that do not affect each other and all come from the same distribution
Exponential distribution	A right-skewed continuous distribution of waiting times between random events
Gamma distribution	A family of right-skewed continuous distributions with a shape and a scale parameter